

4 **(short)** A rigid vessel contains two closed systems of two ideal gases separated by a piston held in position by a pin, as shown in Fig. 4. Initially gas A and gas B have equal volumes of $V_1 = 1 \text{ m}^3$, and gas A has twice the pressure of gas B, $p_1 = 2 p_B$. The pin holding the piston is released. The pressures of both gases are equal at the final state and gas A has a final volume V_2 . The piston is thermally conductive and both systems have a final temperature that is unchanged, $T_2 = T_1$.

(a) Determine whether the heat transfer and work from the initial to the final state are positive or negative for systems A and B. [4]

(b) Calculate V_2 . [6]

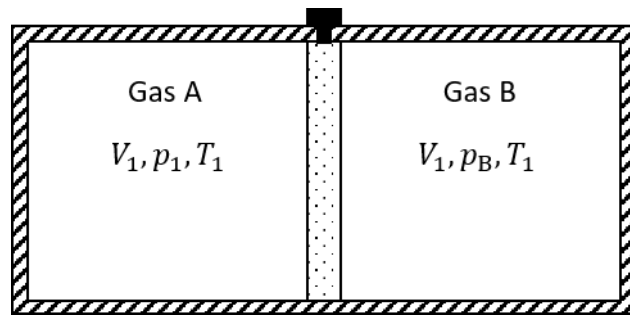


Fig. 4